

SWIFT CURRENT CDA, SK, Canada

WMO: 714460

Lat: 50.260N

Lon: 107.737W

Elev: 2707

StdP: 13.31

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-16.5	-28.0	1.3	-22.0	-21.8	1.8	-16.3	35.3	23.4	31.4	20.0	11.2	280	0.777

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	88.2	63.6	84.3	62.5	80.5	61.4	67.8	80.3	65.4	77.8	63.4	75.9	12.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	98.0	72.9	61.2	89.3	69.9	58.9	82.2	68.1	33.8	80.3	31.8	77.6	30.2	75.7	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
29.3	26.0	23.4	DB	-28.4	94.1	5.1	3.9	-32.0	96.9	-35.0	99.1	-37.9	101.3	-41.6	104.1	
			WB	-28.6	71.6	5.0	3.1	-32.2	73.8	-35.2	75.6	-38.0	77.3	-41.7	79.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	13.2	15.3	25.8	40.0	50.9	59.1	65.6	64.6	55.0	41.4	26.9	15.6
	DBStd	21.96	15.81	14.26	14.55	10.15	8.56	6.46	5.92	6.95	9.06	9.92	13.11	14.44
	HDD50	5393	1140	972	752	320	95	7	0	1	49	295	697	1065
	HDD65	9484	1605	1392	1216	750	441	196	65	95	319	731	1144	1530
	CDD50	1596	0	0	1	21	124	281	485	453	199	29	3	0
	CDD65	212	0	0	0	0	5	20	85	82	20	0	0	0
	CDH74	2749	0	0	0	9	114	272	976	1057	312	9	0	0
	CDH80	910	0	0	0	1	18	68	326	397	100	0	0	0
Wind	WSAvg	12.5	13.9	12.8	13.2	13.2	13.0	12.0	10.7	10.7	11.6	12.6	12.9	13.5
Precipitation	PrecAvg	15.3	0.9	0.7	0.8	1.1	1.9	2.8	1.9	1.7	1.4	0.7	0.6	0.9
	PrecMax	22.6	2.0	1.6	2.8	3.5	5.2	8.7	4.3	5.3	3.9	2.7	1.8	2.2
	PrecMin	10.4	0.1	0.2	0.1	0.2	0.3	0.8	0.3	0.2	0.1	0.0	0.1	0.4
	PrecStd	3.1	0.5	0.4	0.7	0.7	1.2	1.7	1.1	1.2	1.1	0.6	0.4	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.2	45.3	59.9	73.5	82.4	86.9	92.6	93.3	88.3	75.1	61.4	46.5
		MCWB	38.3	38.7	46.9	52.4	57.5	63.7	65.4	63.7	60.2	55.2	47.6	37.8
	2%	DB	39.7	40.2	53.6	66.9	77.1	81.1	87.4	89.0	82.7	67.1	52.7	41.2
		MCWB	35.3	35.5	43.1	49.7	56.4	61.6	66.1	62.7	58.7	51.4	43.3	36.1
	5%	DB	37.2	36.9	48.2	61.7	72.4	76.5	83.4	84.6	76.6	61.0	47.5	37.8
		MCWB	33.9	33.5	40.4	46.7	53.5	59.3	65.3	61.3	56.8	48.5	39.9	33.8
	10%	DB	34.2	33.7	43.1	56.7	67.5	72.6	79.3	80.0	70.9	56.0	42.4	34.3
		MCWB	31.6	31.2	37.4	44.5	51.6	57.7	63.4	60.8	54.5	46.1	36.8	31.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	39.3	47.9	54.2	60.7	69.3	73.3	68.4	62.7	56.1	48.2	39.8
		MCDB	42.4	44.3	58.1	68.4	75.8	80.2	83.2	82.8	78.9	71.9	58.8	44.5
	2%	WB	35.8	35.9	44.0	50.9	57.5	63.9	69.1	65.8	60.2	52.5	43.9	36.4
		MCDB	39.2	39.2	52.4	64.7	73.0	74.5	80.5	80.0	77.6	64.8	51.6	40.5
	5%	WB	33.9	33.7	40.7	48.0	55.2	61.4	66.8	63.9	57.9	49.4	40.4	33.9
		MCDB	37.2	36.6	47.7	60.0	69.7	72.6	78.7	78.4	73.6	59.2	46.9	37.7
	10%	WB	31.5	31.2	37.5	45.2	52.9	59.3	64.7	62.2	55.7	46.6	37.2	31.2
		MCDB	34.2	33.7	42.8	55.4	65.0	69.5	76.1	76.3	69.0	55.0	41.9	34.4
Mean Daily Temperature Range	5% DB	MDBR	15.9	15.8	16.9	20.5	22.2	19.8	22.7	23.8	22.3	19.3	15.8	15.0
		MCDBR	17.0	17.1	23.1	28.8	29.3	26.5	28.5	31.3	30.8	27.1	21.7	17.1
		MCWBR	14.0	13.5	14.6	15.4	13.7	12.1	12.6	12.0	13.6	14.9	14.0	13.1
	5% WB	MCDBR	16.7	16.7	22.1	26.5	25.9	22.6	24.7	26.0	27.6	25.2	21.1	16.9
		MCWBR	14.1	13.5	14.5	15.0	12.8	11.9	12.7	11.8	12.9	14.6	13.9	13.3
Clear-Sky Solar Irradiance	taub	0.245	0.246	0.276	0.344	0.364	0.358	0.364	0.384	0.333	0.286	0.256	0.234	
	taud	2.420	2.488	2.502	2.318	2.306	2.384	2.380	2.288	2.405	2.527	2.499	2.444	
	Ebn at Noon	262	294	300	285	283	285	281	269	273	272	257	251	
	Edh at Noon	19	23	27	37	39	36	36	37	30	22	17	16	
All-Sky Solar Radiation	RadAvg	386	705	1127	1537	1845	1954	2121	1711	1250	768	420	297	
	RadStd	41	54	81	109	131	133	129	129	112	88	46	29	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	-2.34	-3.04	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page